Operations Research 1: Deterministic Models
Python Setup

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Terms

Terms

- Python
  - An easy but powerful programming language
  - Has an active community of contributors, providing thousands of packages that you can use
- Conda
  - An open-source package manager and environment management system
  - This course will not cover any content in Conda.
- Anaconda
  - A distribution of Python for scientific computing that aims to simplify package management using Conda

Source: https://www.anaconda.com/, https://docs.conda.io/en/latest/

Terms

- Spyder
  - An open-source integrated development environment (IDE) for scientific programming in the Python language
  - Used to write some complex programs like formulating a linear program in this course
- Jupyter Notebook
  - An open-source web application that allows you to create and share documents that contain live code, equations, visualizations and narrative text
  - More like a textbook for Python and an interactive window for manipulating data in this course

Source: https://www.anaconda.com/open-source

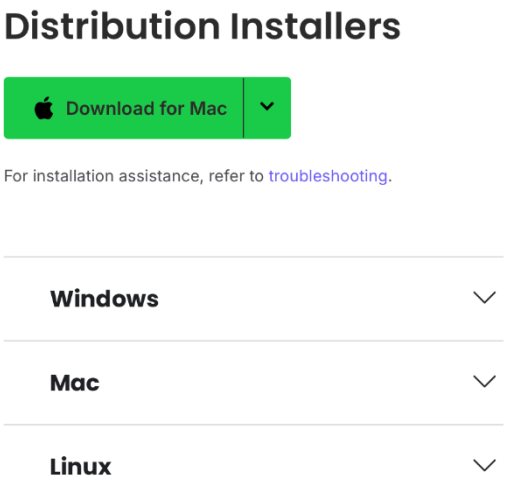
Environment Setup

Environment Setup

- Notes:
  - You are free to set up your Python environment using any package management tool (e.g., Pyenv, Conda) and any IDE (e.g., VS Code, PyCharm) for coding, based on your preference. This course does not require a specific setup.
  - This tutorial only provides an example on how to build a Python environment with Anaconda.

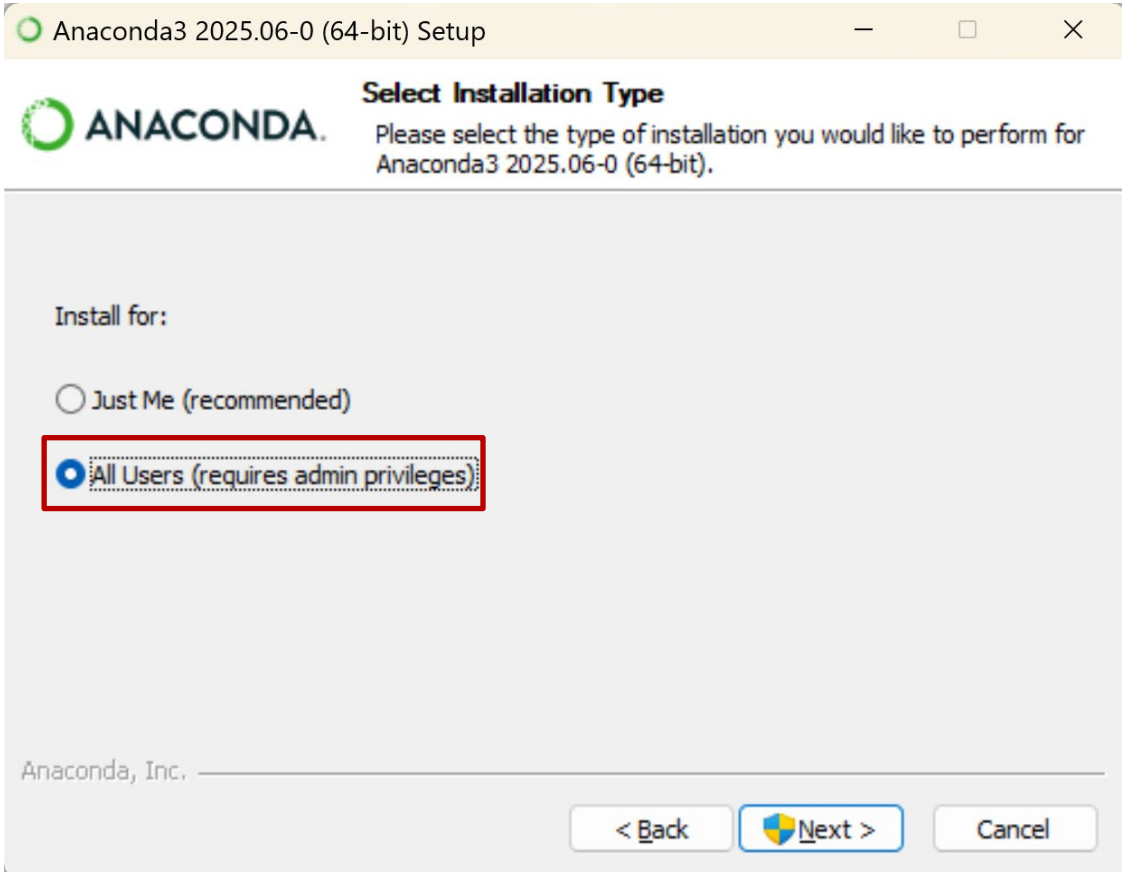
Environment Setup

- Download the latest version of Anaconda.
  - https://www.anaconda.com/download/success



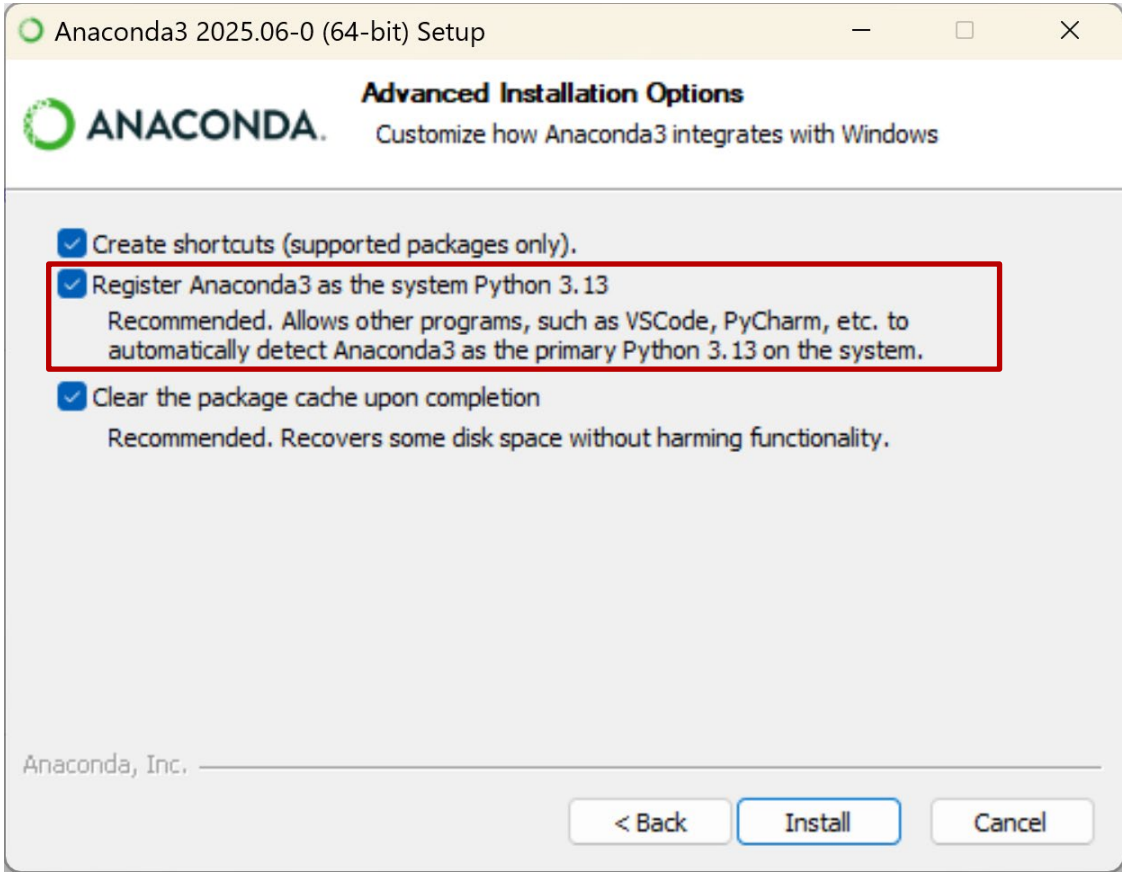
Environment Setup

- Install Anaconda (for Windows OS)
  - The default setting is fine for beginners.



Environment Setup

- Install Anaconda (for Windows OS)
  - The default setting is fine for beginners.



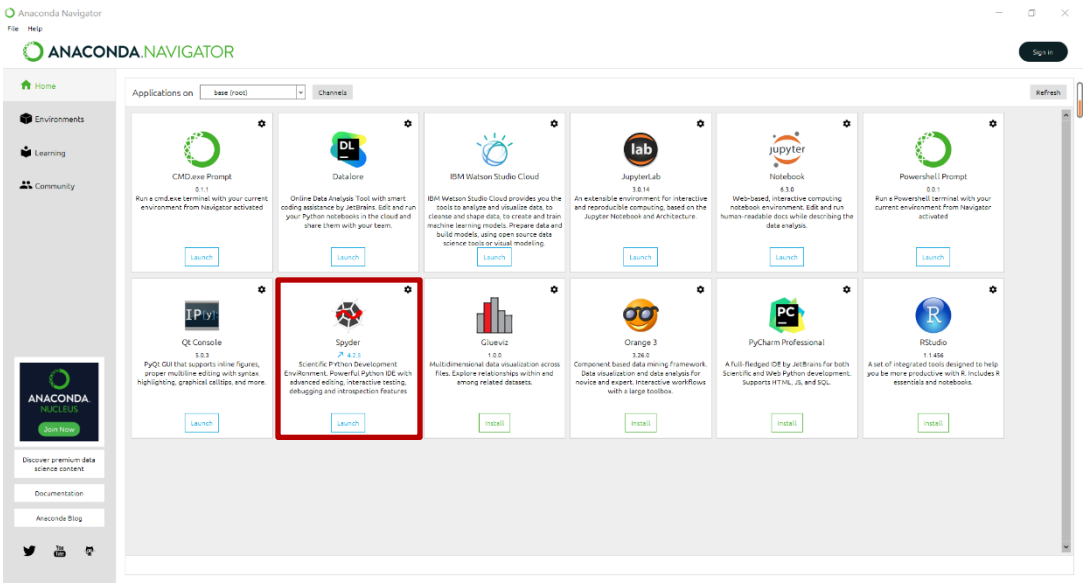
Environment Setup

- Install Anaconda (for Mac OS)
  - The default setting is fine for beginners.

First Python Program

First Python Program

- Open Spyder using Anaconda Navigator.

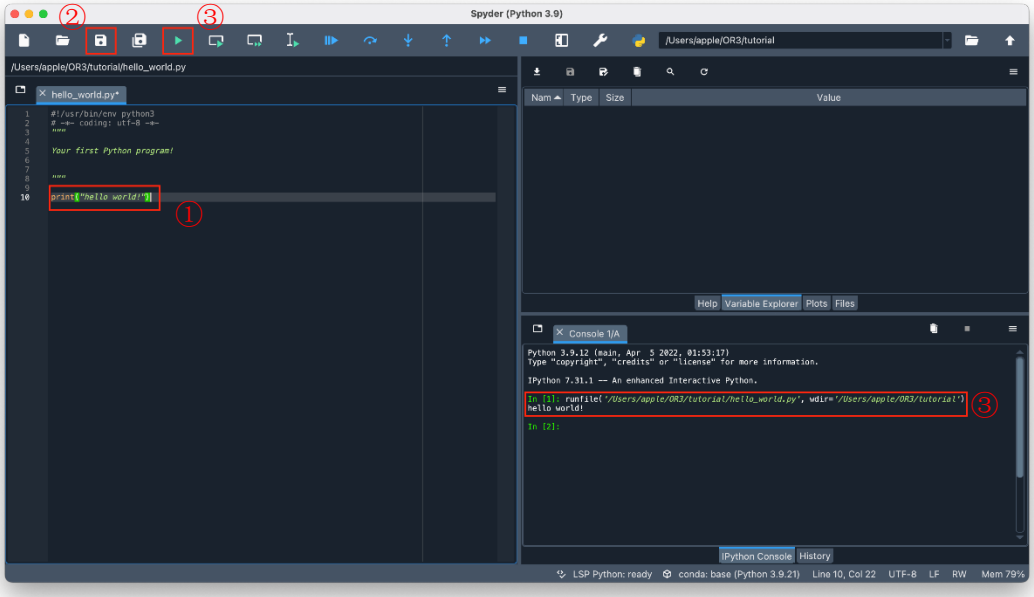


First Python Program

- Enter the following code in Spyder.

```
print("Hello, world!")
```

- Save the program to your working directory.
  - "File → Save"
  - Name the program file, such as "hello_world.py."
- Run the program.

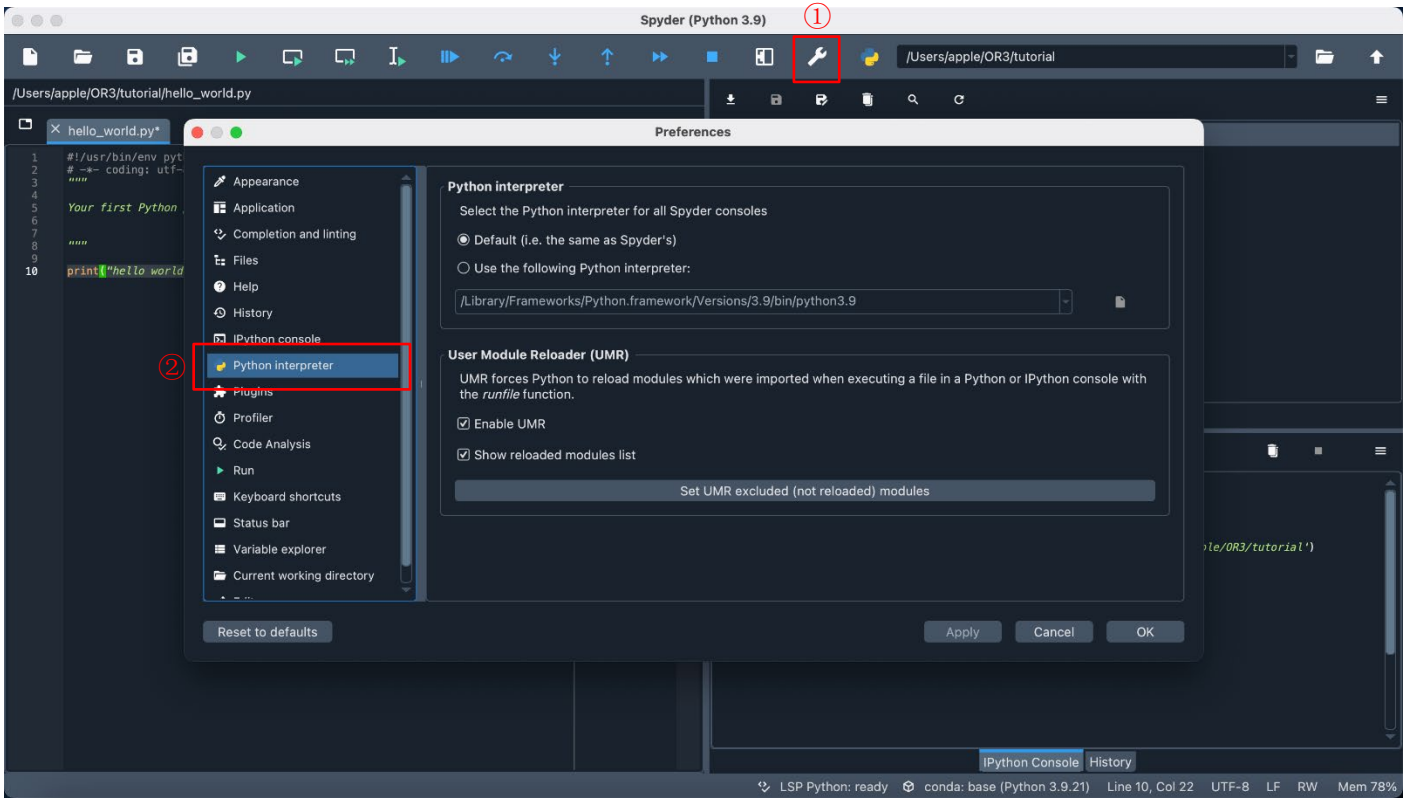


First Python Program

- Change Python interpreter
  - If a “package not found” error is raised even after installation, the package was likely installed in a different Python interpreter and changing it may resolve the problem.
  - Simply use the interpreter list to select and activate the appropriate environment for your current project, maintaining an organized workspace with the proper package versions.

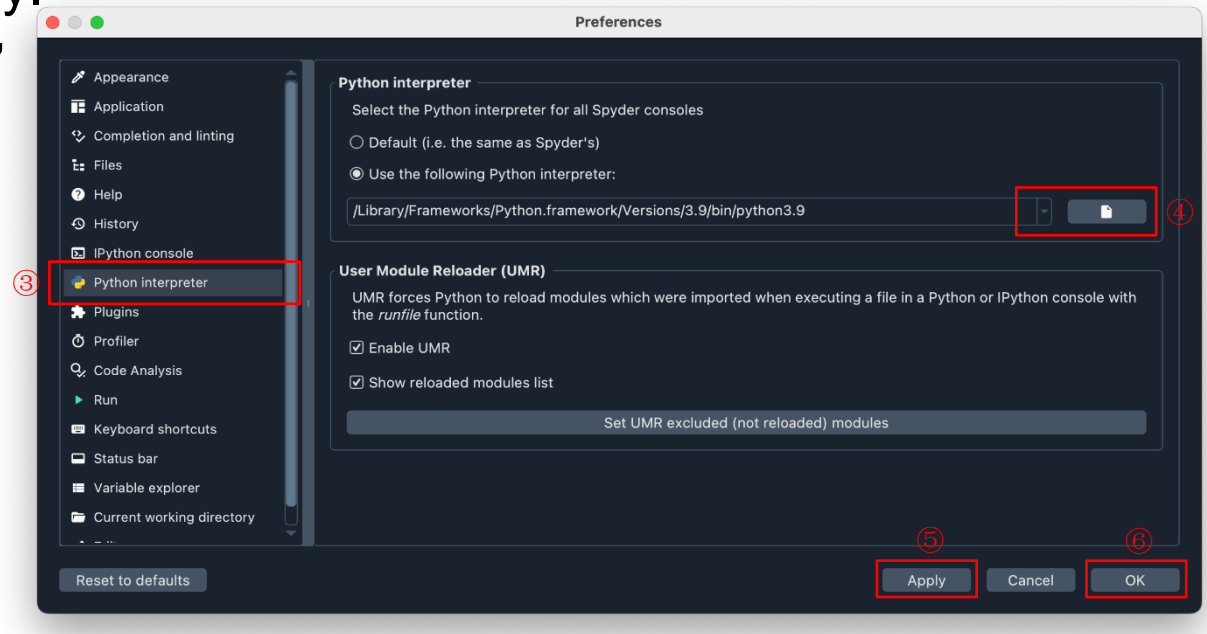
First Python Program

- Change Python interpreter
  - ① Open “Preferences.”
  - ② Select “Python interpreter.”



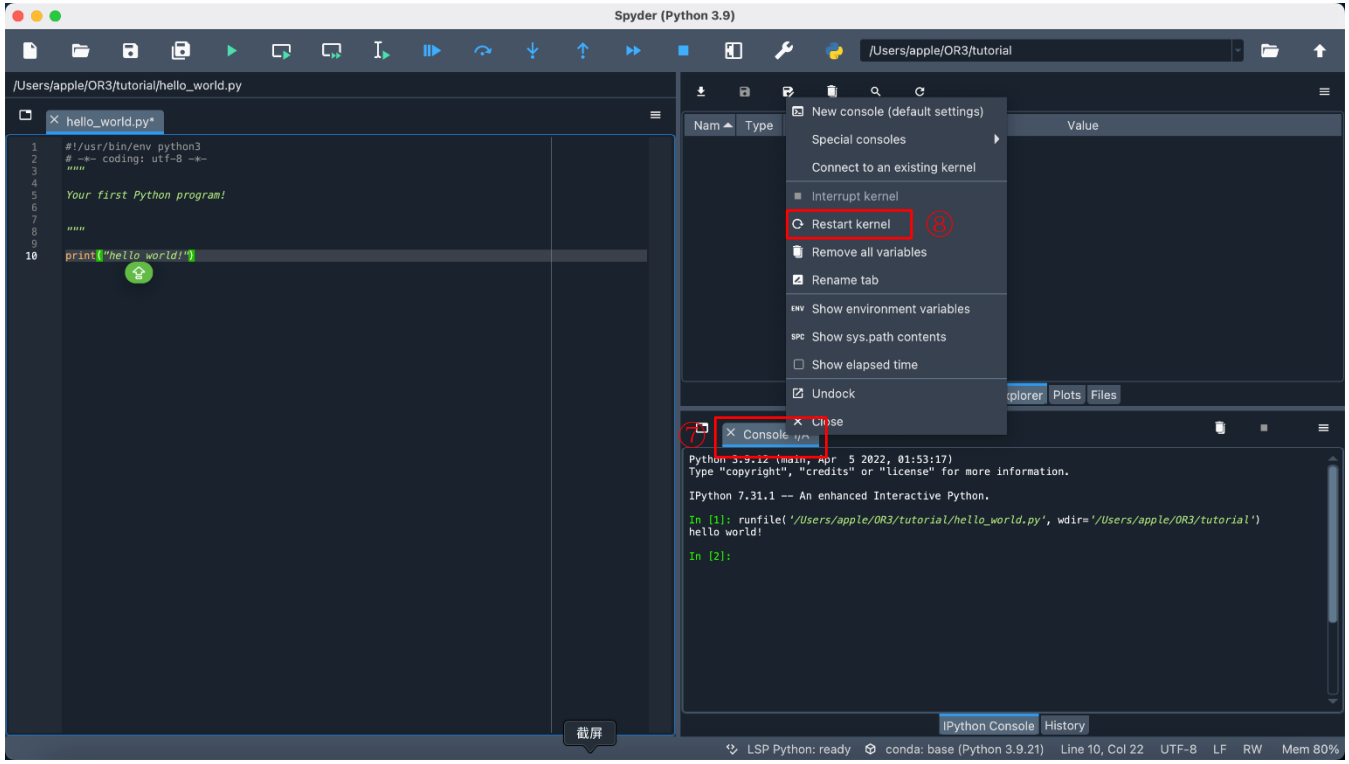
First Python Program

- Change Python interpreter
  - ③ Select “Use the following Python interpreter.”
  - ④ From the interpreter drop-down list, select your desired interpreter. If the correct one is not listed, click the “select file” button to browse and add it.
  - ⑤ Select “Apply.”
  - ⑥ Select “OK.”



First Python Program

- Change Python interpreter
  - ⑦ Find “IPython Console.”
  - ⑧ Right-click to select “Restart kernel.”



Further Information

Python Self-Study References

- This course is NOT a crash course on Python.
- Below are some references for self-study.
  - Python documentation: https://docs.python.org/3
    - Use this as a reference, but not a textbook.
  - Textbook: Matthes, E. (2019). Python Crash Course, 2nd Edition: A Hands-On, Project-Based Introduction to Programming. No Starch Press.
  - CS61A at UC Berkeley: https://cs61a.org
    - An open access course for beginners in programming using Python
- You may use any references that best suit you.